FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Malaria in Children

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Namugga

Plasmodium falciparum malaria is the leading cause of illness and death in Ugandan under-fives; every febrile child gets a malaria test, uncomplicated malaria gets a weight-banded ACT, and severe malaria (any one danger feature) gets IV or IM artesunate at once, with glucose, fluids, blood and seizures managed as part of the same emergency.

Must know before the ward round

1. Test before you treat (UCG 2023): every febrile child gets an RDT or blood slide. An HRP2 RDT stays positive for 2–3 weeks after cure, so use a blood slide if fever returns within 3 weeks.
2. Uncomplicated malaria: artemether–lumefantrine (20/120 mg) by weight: 5–14 kg 1 tab, 15–24 kg 2, 25–34 kg 3, ≥35 kg 4; six doses at 0, 8, 24, 36, 48 and 60 h with milk or food.
3. Severe malaria = parasitaemia plus ANY one of: impaired consciousness/coma, prostration, ≥2 convulsions in 24 h, respiratory distress/acidosis, Hb <5 g/dL, glucose <2.2 mmol/L, shock, haemoglobinuria, jaundice, bleeding, renal failure, hyperparasitaemia.
4. IV artesunate: 3 mg/kg if under 20 kg, 2.4 mg/kg if 20 kg or more, at 0, 12 and 24 h, then daily until the child can take oral drugs; then a full 3-day ACT (Uganda: DHA–piperaquine).
5. Pre-referral rectal artesunate (children under 6 only): 100 mg (1 suppository) for 4 months to <3 years; 200 mg (2 suppositories) for 3 to <6 years. Refer at once: rectal artesunate alone is not a cure.
6. Blantyre coma scale (max 5): eyes 0–1, best motor 0–2, best verbal 0–2. A score of 2 or less = unrousable coma (cerebral malaria) once hypoglycaemia and a post-ictal state are excluded.
7. Hypoglycaemia: give 10% dextrose 5 mL/kg IV over 5–7 min (1 mL/kg of 50% dextrose + 4 mL/kg water), recheck in 15–30 min. Never give undiluted 50% dextrose to a child.
8. Transfuse whole blood 20 mL/kg (or packed cells 10–15 mL/kg) over 3–4 h if Hb ≤4 g/dL, or Hb 4–6 g/dL with respiratory distress, shock, impaired consciousness or heart failure.
9. No routine fluid boluses (FEAST trial): give boluses only for true shock; otherwise maintenance fluids or NGT feeds, and watch the liver edge for overload.
10. Convulsions: diazepam 0.5 mg/kg rectally or 0.2–0.3 mg/kg slow IV (max 10 mg), once more after 10 min; then phenobarbital 15 mg/kg IV/IM loading. Check glucose every time.
11. Uganda added the malaria vaccine to routine immunisation in April 2025: 4 doses at 6, 7, 8 and 18 months in 107 high-burden districts.
12. Artemisinin partial resistance (kelch13 mutations) has been documented in northern Uganda, so check that parasites are clearing and always complete the partner drug.

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1. Definition

Malaria is an acute febrile illness caused by *Plasmodium* parasites carried by female *Anopheles* mosquitoes. In Uganda, **P. falciparum** causes almost all cases and almost all deaths. The Ministry of Health still calls malaria **the leading cause of illness and death in young children**. It fills paediatric wards at Mengo and Mulago and every HC IV in the country.

What you have to grasp is that malaria in a child is either **uncomplicated**, treated with a 3-day oral ACT at home, or **severe**, a medical emergency treated with **parenteral artesunate** plus management of each complication. A child can go from one to the other **within 24 hours**. Most deaths come from **cerebral malaria, severe anaemia with respiratory distress (acidosis), hypoglycaemia** and delays in referral.

> CLINICAL PEARL
> The one thing never to forget: **in a sick febrile child, check three things in the first 5 minutes: consciousness (Blantyre score), blood glucose and Hb.** Then give the first dose of artesunate. Coma, hypoglycaemia and severe anaemia kill within hours, and all three can be treated.

Term	Definition (UCG 2023 / WHO)
Uncomplicated malaria	Symptomatic malaria (fever ≥37.5 °C axillary or a history of fever) with a positive RDT or blood slide and **no** feature of severe malaria
Severe malaria	Asexual *P. falciparum* in blood **plus at least one** severe feature (section 5), with no other confirmed cause
Cerebral malaria	Unrousable coma (**Blantyre ≤2**, or unable to localise pain) with parasitaemia, lasting **>1 hour after a seizure**, with hypoglycaemia corrected and other causes (meningitis) excluded
Severe malarial anaemia	**Hb <5 g/dL** (packed cell volume <15%) with parasitaemia (UCG 2023; WHO adds parasite count >10,000/µL)
Hyperparasitaemia	Parasite density **>10%** of red cells, or >250,000/µL (UCG)
Treatment failure	Parasites return or persist after a full, correctly taken course (blood slide, not RDT)

2. Epidemiology

- **Global (WHO 2025 fact sheet, 2024 data):** about **282 million cases and 610,000 deaths**; the WHO African Region carries **95%** of both, and **children under 5 account for about 75% of malaria deaths** in the region.
- **Uganda** is consistently among the countries with the highest number of cases worldwide. Transmission is **perennial and high** in most districts (West Nile, Acholi, Lango, Busoga, Karamoja, the east). It is lower and **epidemic-prone in the highlands** (Kabale, Kisoro, Kasese uplands), where older children with little immunity get severe disease.
- **Age**: in high-transmission areas, severe malaria hits the children with the least immunity: **6 months to 5 years**. Severe anaemia peaks at 6–24 months, cerebral malaria at 2–5 years. Babies under 6 months are partly protected by maternal antibody and fetal haemoglobin.

UGANDA CONTEXT

The care pathway in Uganda: VHTs (iCCM) and HC IIs test with RDTs and give AL. They give **rectal artesunate before referral** for danger signs. HC IIIs give AL and IM/IV artesunate before referral and admit some cases. HC IVs and hospitals manage severe malaria with **IV artesunate, glucose, blood transfusion** (only where there is a blood bank supply) and monitoring. **Blood is the commonest missing resource** at HC IV level, and severe anaemia is often a reason to refer. Mengo, Mulago, Lacor and the regional referral hospitals (RRHs) have paediatric emergency units, the "Acute Care Unit" model.

UGANDA CONTEXT

Artemisinin partial resistance is in Uganda. Parasites with **kelch13 mutations** (e.g. A675V, C469Y) and slow clearance after artesunate were documented in **northern Uganda** (Balikagala et al., NEJM 2021). The partner drug (lumefantrine, piperaquine) still cures most infections. In practice: **give the full 3-day course, give the partner drug properly with food, and use a blood slide to check that parasites are clearing in children who do not improve.**

3. Aetiology and Risk Factors

Cause: *Plasmodium falciparum* (almost all Ugandan cases and deaths); *P. vivax*, *P. ovale* (dormant liver hypnozoites, relapse, need primaquine) and *P. malariae* are uncommon. Vector: female *Anopheles* mosquitoes, biting mainly at night.

Risk factors for severe disease

Risk factor	Why it matters
Age 6 months–5 years in high-transmission areas	Least acquired immunity
Non-immune child (highlands, or travel between highland and lowland)	Severe disease at any age; adult-type picture
Sickle cell disease	High mortality with malaria; crises and severe anaemia
HIV	More severe malaria, other infections
Severe acute malnutrition	Changes fluids, transfusion volume and glucose threshold
No net	Exposure
Missed or incomplete ACT course	Treatment failure, progression
Herbal or "local" treatment first	Delays care
Iron deficiency, hookworm	Worsen anaemia
G6PD deficiency + oxidant drugs	Haemoglobinuria

4. Pathophysiology

Life cycle in one paragraph. An infected *Anopheles* injects **sporozoites**. These invade liver cells and multiply silently for about **7–10 days** (the incubation period is typically 10–15 days). Then **merozoites** burst into the blood and invade red cells: ring → trophozoite → **schizont**. Each 48-hour cycle ruptures red cells and releases new merozoites; the burst causes the fever. Some parasites become **gametocytes**, which infect the next mosquito. *P. vivax* and *P. ovale* leave dormant liver **hypnozoites** (relapse; need primaquine), but they are uncommon in Uganda.

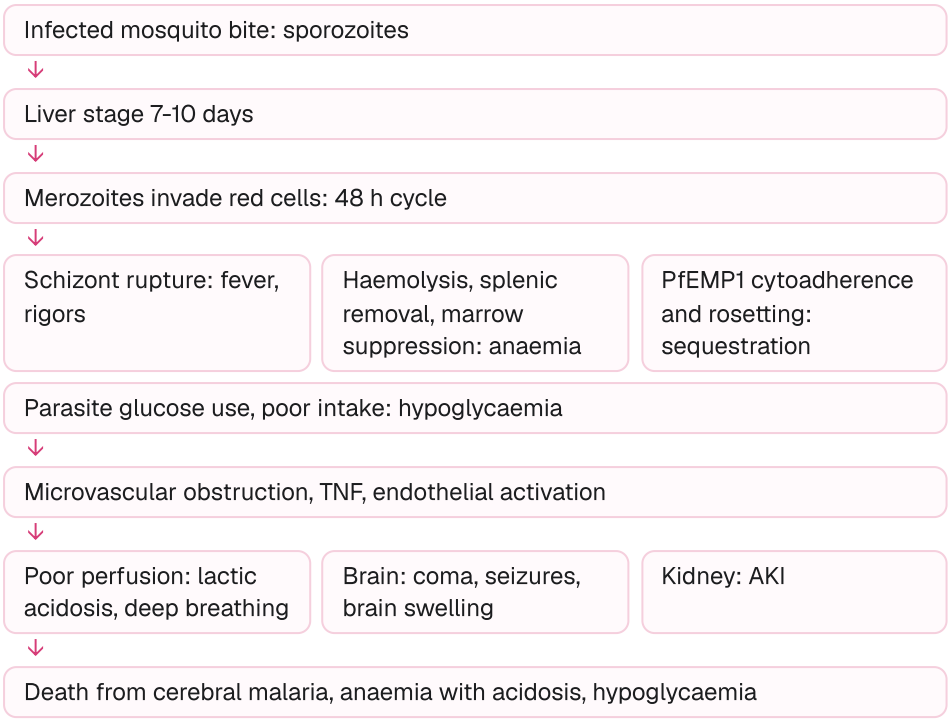
Why *P. falciparum* kills: sequestration

- Infected red cells express **PfEMP1** on surface "knobs". They **stick to the endothelium** of small vessels (cytoadherence) and to uninfected cells (rosetting). Mature parasites therefore **hide in the brain, gut, kidney and placenta** and are not seen on the peripheral film. This is why the **peripheral parasite count can underestimate** the total burden.
- **Microvascular obstruction + inflammation (TNF) + endothelial activation** → reduced tissue perfusion → **lactic acidosis**, coma, renal injury.
- **Brain swelling** occurs in many children who die of cerebral malaria (shown on MRI in Malawian children). This is why **deep coma with abnormal posturing and irregular breathing** is so dangerous.

Mechanisms of each complication

Complication	Mechanism	So you will…
Anaemia	Rupture of parasitised cells, **removal of uninfected cells** by the spleen, **bone-marrow suppression** (dyserythropoiesis); worse with iron deficiency, hookworm, HIV, sickle cell	Transfuse when Hb is very low or the child is decompensating; treat iron deficiency later
Respiratory distress	Mostly **metabolic acidosis** (lactate from poor perfusion and anaemia), not pneumonia	Correct anaemia and perfusion; the deep breathing settles after transfusion
Hypoglycaemia	Children have small glycogen stores, fast during illness, and parasites consume glucose; **quinine stimulates insulin release**	Check glucose on arrival, 4-hourly and after every seizure; feed by NGT
Seizures	Sequestration, fever, hypoglycaemia, hyponatraemia; may be subtle (eye deviation, twitching)	Treat seizures; look for subtle ones in a child who stays in coma
Haemoglobinuria / blackwater	Intravascular haemolysis (also G6PD deficiency + oxidant drugs)	Hydrate, check Hb and renal function
Delayed haemolysis after artesunate (PADH)	Red cells that had parasites "pitted" out by the spleen die 1–3 weeks later	Recheck Hb at about 7, 14 and 28 days after severe malaria

Why artesunate beats quinine. Artemisinins kill **young ring stages** before they mature and sequester, so fewer parasites reach the microcirculation. In **AQUAMAT** (5,425 African children, Lancet 2010), artesunate reduced mortality from **10.9% to 8.5%** (a 22.5% relative reduction) and caused fewer convulsions, less deep coma and less hypoglycaemia than quinine.



Feature links

- Schizont rupture and cytokine release (TNF) → fever, rigors, headache, vomiting, anorexia.
- Red cell destruction + splenic clearance + marrow suppression → pallor, severe anaemia (Hb <5 g/dL), tachycardia, heart failure (gallop, enlarging liver).
- Anaemia + poor perfusion → lactic acidosis → deep (Kussmaul) breathing, nasal flaring, indrawing without crackles.
- Splenic hyperplasia → splenomegaly.
- Intravascular haemolysis → haemoglobinuria (dark, "Coca-Cola" urine with no red cells); haemolysis and liver dysfunction → jaundice.
- Cerebral sequestration + brain swelling → impaired consciousness, coma, abnormal posturing, irregular breathing, unequal pupils; retinal sequestration → malarial retinopathy.
- Sequestration, fever, hypoglycaemia, hyponatraemia → convulsions, including subtle seizures (eye deviation, twitching).
- Small glycogen stores + fasting + parasite consumption + quinine-induced insulin release → hypoglycaemia → drowsiness, seizures, coma.
- Reduced renal perfusion → oliguria (<0.3 mL/kg/h), raised creatinine.
- Endothelial activation, thrombocytopenia, DIC → bleeding from gums, nose, venepuncture sites.
- Hypovolaemia, sepsis (NTS) → shock: cold hands, CRT >3 s, weak fast pulse.
- Prostration → cannot sit or drink: a severe feature.
- Pitting of once-infected red cells after artesunate → delayed haemolysis 1–3 weeks later.

5. Classification / Staging

Severe malaria features (UCG 2023, adapted for children). Any **one** feature with parasitaemia = severe malaria.

Feature	Criterion
Impaired consciousness / coma	Blantyre score <3 = coma; confusion, drowsiness or abnormal behaviour = impaired consciousness

Feature	Criterion
Prostration	Cannot sit unsupported (if normally able to), or cannot drink or breastfeed if too young to sit
Multiple convulsions	**≥2 convulsions in 24 h**
Respiratory distress / acidosis	Deep (Kussmaul) breathing, nasal flaring, chest indrawing; bicarbonate <15 mmol/L; lactate ≥5 mmol/L
Severe anaemia	Hb **<5 g/dL**
Hypoglycaemia	Glucose **<2.2 mmol/L** (40 mg/dL) in UCG; WHO Pocket Book treats below **2.5 mmol/L** (<3 mmol/L if severely malnourished)
Shock (circulatory collapse)	Cold peripheries, capillary refill >3 s, weak fast pulse; UCG cites systolic BP <50 mmHg in children
Renal impairment	Urine output <0.3 mL/kg/h (UCG: <12 mL/kg in 24 h) with raised creatinine
Haemoglobinuria	Dark ("Coca-Cola") urine with no red cells on microscopy
Jaundice	Clinical jaundice with parasitaemia
Abnormal bleeding	Gums, nose, venepuncture sites, haematemesis, melaena
Pulmonary oedema	Rare in children; more common in adults
Hyperparasitaemia	>10%

Children have three main **syndromes**: **cerebral malaria**, **severe malarial anaemia** and **respiratory distress (metabolic acidosis)**. They often overlap, and the overlap carries the highest mortality.

Blantyre coma scale (children too young to speak; max 5)

Domain	Response	Score
Eye movement	Watches or follows (e.g. mother's face)	1
	Fails to watch or follow	0
Best motor response (painful stimulus: firm pressure on sternum or nail bed)	Localises the pain	2
	Withdraws the limb	1
	Non-specific or absent response	0
Best verbal response	Cries appropriately with pain (or speaks, if verbal)	2
	Moan or abnormal cry	1
	No vocal response	0

Interpretation: 5 = normal; 3–4 = impaired consciousness (still a severe feature); **≤2 = unrousable coma**. Record it on arrival and every 4 hours (more often if it falls). Use GCS for older children.

6. Clinical Features

History

Question	Why it matters
Fever: how long, pattern, rigors?	Malaria usually <7 days; fever >7 days → think typhoid, TB, UTI, abscess, HIV, leukaemia

Question	Why it matters
Convulsions? How many in 24 h, how long, focal?	≥2 in 24 h = severe malaria; a long or focal seizure → think meningitis or encephalitis as well
Level of activity: playing, drowsy, can't sit, can't drink or breastfeed?	**Prostration and inability to drink** are severe features (IMCI danger signs)
Vomiting everything?	Cannot take oral ACT → needs parenteral treatment
Dark urine, reduced urine, yellow eyes	Haemoglobinuria, AKI, jaundice
Fast or difficult breathing	Acidosis or pneumonia (check both)
Treatment already given: which antimalarial, dose, how many days, herbal remedies?	Partial treatment, incomplete course (failure); quinine within 12–24 h changes the quinine loading dose
Recent travel (highland child visiting the lowlands, or the reverse); where the child lives	Non-immune → severe disease at any age
Sickle cell disease, HIV status (child and mother), cotrimoxazole, TB contact	SCD → crises and severe anaemia; HIV → more severe malaria, other infections; other diagnoses
Immunisation (UNEPI card incl. malaria vaccine at 6, 7, 8, 18 months), **vitamin A, deworming**, **net use last night**	Prevention counselling; vaccine eligibility; missed doses
Feeding and growth (weight-for-age, MUAC), recent weight loss	SAM changes fluids (no boluses), transfusion volume (10 mL/kg) and the glucose threshold
Previous admissions, transfusions	Recurrent severe anaemia → post-discharge chemoprevention, look for SCD

Examination

ETAT triage comes first (section 9). Then follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, leave distressing parts (fundoscopy, ears, throat) to last, and judge vital signs against age norms.

Introduction, consent and preparation

- Introduce yourself; confirm identity; explain; consent from the parent and assent from the child; **hand hygiene**; position on the parent's lap (recovery position if unconscious); expose as needed in a warm room.

General inspection (end of the bed)

Sign	Mechanism	What it means
Prostrate, cannot sit or drink	Severe systemic illness	Severe malaria
Drowsy, unrousable, posturing	Cerebral sequestration, brain swelling	Cerebral malaria (after excluding glucose, seizures, meningitis)
Deep (acidotic) breathing	Lactic acidosis	Severe anaemia with acidosis: give blood
Subtle seizures (eye deviation, twitching)	Ongoing seizure activity	Treat; a reason for persistent coma
Severe palmar pallor (palms white)	Severe anaemia	IMCI sign of severe anaemia
Jaundice; bleeding	Haemolysis, liver dysfunction; DIC	Severe features
Wasting, bilateral pedal oedema	SAM (kwashiorkor)	SAM protocol

Vital signs and growth: RR, HR, CRT, BP, SpO_2, temperature, **weight** (needed for every dose), MUAC; plot weight-for-age and height-for-age.

Normal vital signs by age (guide; use the measured weight for every dose)

Age	Respiratory rate (/min)	IMCI "fast breathing"	Heart rate (/min)	Systolic BP (mmHg)
<2 months	30–60	**≥60**	110–170	60–80
2–12 months	30–40	**≥50**	110–160	70–90
1–5 years	25–35	**≥40**	95–150	80–100
5–12 years	20–25	not defined by IMCI (use age norms)	80–120	90–110

Quick lower limit of normal systolic BP for 1–10 years: **70 + (2 × age in years)**.

Hands, face and neck

- **Hands**: cold peripheries, CRT >3 s (shock); palmar pallor.
- **Face**: conjunctival and tongue pallor; scleral jaundice; bleeding gums; eyelid oedema (fluid overload).
- **Neck**: neck stiffness (meningitis); in infants, a bulging fontanelle.

The systems

- **Respiratory**: rate, **deep breathing**, indrawing, crackles (pneumonia vs acidosis), SpO_2.
- **Cardiovascular**: pulse, CRT, gallop, **liver size** (baseline for overload; hepatomegaly suggests heart failure or fluid overload).
- **Abdomen**: spleen (in cm below the costal margin; malaria or SCD sequestration), liver, bladder.
- **Neurological**: **Blantyre or GCS**, pupils, posturing (decorticate, decerebrate), tone, eye deviation, neck stiffness, fontanelle.

Completing the examination

- **Look for another focus**: chest, ears, throat, skin, joints, urine (colour, dipstick).
- **Retina, last**, if you can: malarial retinopathy supports cerebral malaria.
- **Thank**, **wash hands**, **summarise** with the severe features present.
- **Document negatives**: "no neck stiffness, fontanelle not bulging, chest clear, no jaundice".

Red flags

RED FLAG

- **Blantyre ≤2**, or a fall of 1 point or more; abnormal posturing; irregular breathing; unequal pupils (raised ICP)
- **Deep (acidotic) breathing**, with pallor = severe anaemia with acidosis: transfuse urgently
- **Glucose <2.5 mmol/L**, or any child in coma or convulsing whose glucose has not been measured
- **Shock**: cold hands, CRT >3 s, weak fast pulse
- Repeated convulsions, or a seizure lasting more than 5 minutes
- **Haemoglobinuria**, anuria, jaundice with bleeding
- SAM, sickle cell disease or a child under 6 months with malaria

Thinking like a clinician

- **A positive RDT does not mean malaria is the whole story.** In high-transmission areas many children carry parasites. A child with a positive RDT and a stiff neck still needs an LP. One with a positive RDT and a high respiratory rate with crackles still needs pneumonia treatment. **Co-infection with non-typhoidal *Salmonella* bacteraemia** is common in African children with severe malaria, especially with anaemia, so a very sick child gets **antibiotics too**.

- **Coma + positive RDT** → correct glucose, stop seizures, then reassess after 1 hour. If coma persists, think cerebral malaria **and** meningitis: do an LP if it is safe, or give ceftriaxone if it is not.
- **Deep breathing + very pale** = **severe anaemia with acidosis**: the treatment is **blood**, not oxygen alone or a fluid bolus.
- **Fast breathing + crackles + fever** → may be pneumonia **or** acidosis; check the Hb and look for indrawing.
- **Fever back within 3 weeks of treatment** → RDT (HRP2) may still be positive from the old infection. Do a **blood slide**, and think of other causes.
- **Highland child or older child with severe malaria** → non-immune: more cerebral malaria, renal failure and jaundice (adult-type picture).
- **Common student mistakes**: treating fever as malaria without a test; forgetting the glucose; giving a bolus to a pale, breathless child; giving the IV artesunate dose for the wrong weight band (3 mg/kg under 20 kg); stopping after the parenteral doses without a full oral ACT; forgetting to recheck the Hb after discharge.

EXAM FAVOURITE

Viva favourite: **"This child has a positive RDT and is in coma. Is it cerebral malaria?"** Model answer: Not yet. Cerebral malaria is a diagnosis of exclusion: **correct hypoglycaemia, stop seizures and wait at least 1 hour** for the post-ictal state to pass, and **exclude meningitis** (LP if safe, or treat empirically). Only persistent unrousable coma (Blantyre ≤2) with parasitaemia and no other cause is cerebral malaria.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Bacterial meningitis	Neck stiffness, bulging fontanelle, focal seizures, purpura; can coexist with a positive RDT	LP (CSF cells, Gram stain, glucose)
Viral encephalitis	Behaviour change, focal signs, seizures; negative or incidental malaria test	LP (lymphocytes, normal glucose)
Septicaemia (incl. NTS)	Shock, very high or low WBC, focus (skin, bone); malnourished or HIV	Blood culture, FBC
Pneumonia	Cough, fast breathing with indrawing, crackles, hypoxia	Clinical; chest X-ray
Typhoid fever	Fever >7 days, abdominal pain, relative bradycardia, rose spots rare	Blood culture (Widal unreliable)
UTI	Infant with fever and no focus, dysuria, smelly urine	Urine dipstick and culture
Sickle cell crisis	Known SCD, dactylitis, pain, sudden anaemia with big spleen (sequestration)	Sickling test, Hb trend, reticulocytes
Hypoglycaemia from another cause / poisoning	History of ingestion (e.g. organophosphates, kerosene), fasting	History, glucose
Febrile convulsion	6 months–5 years, single brief generalised seizure, fully alert within 1 h	Clinical; malaria test still required
Measles / other viral fever	Rash, coryza, conjunctivitis	Clinical
Leukaemia, HIV, TB	Prolonged fever, weight loss, lymphadenopathy, bleeding	FBC and film, HIV test, TB workup

8. Clinical Workup

1. **Never treat without a test** (UCG 2023), but **never delay parenteral artesunate in a child with danger signs** while waiting for a slide. A negative RDT with severe features and no other cause → UCG says treat as severe malaria **with antibiotics** and repeat the RDT in 12–24 h.
2. **RDT vs slide:** use the **blood slide** if the child has completed treatment and is still sick, returns within 3 weeks, or is RDT-negative with no other cause.
3. **Glucose and Hb are the two most useful tests in severe malaria**, and both can be done in minutes on the ward.

"Available at" is the lowest level that usually offers the test Verify *compiled from the source notes, not an MoH list*.

Bedside

Test	Why / how	Expected finding	How it changes management	Available at
Malaria RDT (HRP2 or HRP2/pLDH)	Every febrile child; finger prick; read at 15–20 min	Positive = *P. falciparum* antigen; stays positive 2–3 weeks after cure	Treat; slide if recent treatment	All levels, including VHT
Blood glucose (glucometer)	On arrival, after each seizure, 4-hourly in severe malaria (more often on quinine)	<2.5 mmol/L	Treat now with 10% dextrose	HC III and above Verify
Hb / HemoCue	Finger prick	<5 g/dL = severe malarial anaemia	Guides transfusion	HC III and above Verify
Blantyre coma scale	Section 5	≤2 = coma	Severity and trend	All levels (clinical)
SpO_2	Pulse oximeter	<90%	Oxygen	HC IV and above
Urine colour, dipstick	Inspect; dipstick blood + no RBCs on microscopy	Haemoglobinuria	Hydrate; check Hb, creatinine	HC III and above
Bedside clotting test	If bleeding	Prolonged	DIC: fresh whole blood, vitamin K	Hospitals

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Thick and thin blood film	All admissions; repeat at 24–48 h if not improving	Thick: parasites present; thin: species and **% parasitaemia**; >10% = hyperparasitaemia	Confirms diagnosis; falling count = response	HC III and above
Full blood count	Severe malaria	Low Hb, low platelets (common, rarely needs treatment), high WBC suggests bacterial co-infection	Transfusion, antibiotics	HC IV and above
Group and cross-match	Hb <6 g/dL or falling, bleeding	–	Transfusion	Hospitals and some HC IVs (UBTS supply)
HIV test	Every admitted child with unknown status	Positive	Cotrimoxazole, ART referral	All levels

Test	When	Expected / abnormal finding	How it changes management	Available at
Sickling test	Recurrent severe anaemia, family history, splenomegaly in an older child	Positive	SCD care, prophylaxis	Many hospitals

Imaging

Test	When	Expected finding	How it changes management	Available at
Chest X-ray	Crackles, hypoxia	Consolidation (pneumonia) vs clear (acidosis)	Antibiotics	Hospitals

Specialised / referral-level

Test	When	Expected finding	How it changes management	Available at
Blood culture	Severe malaria, especially with anaemia, shock or high WBC	NTS, *S. pneumoniae*, Gram-negatives	Antibiotic choice	RRH, NRH, Mengo, Lacor
Lumbar puncture	Coma or convulsions, especially with a negative test or coma >1 h after correcting glucose and seizures, if no contraindication	Normal CSF in cerebral malaria (opening pressure may be high)	Excludes meningitis; if not possible, give ceftriaxone	Hospitals
Urea, creatinine, electrolytes	Oliguria, dark urine, deep breathing, older child	Creatinine raised (AKI), sodium low	Fluids, furosemide, referral for dialysis	RRH, NRH
Lactate / blood gas	Deep breathing	Lactate ≥5 mmol/L, bicarbonate <15	Severity; response to transfusion	NRH, teaching hospitals
Hb electrophoresis	Positive sickling test	HbSS	SCD care	Referral labs

Putting it together

Model one-line summary

> "A 3-year-old boy, 12 kg (weight-for-age Z between −1 and −2), fully immunised except for the malaria vaccine (not yet introduced in his district at his age), HIV negative, presented with 3 days of fever and three generalised convulsions today. He has severe malaria with impaired consciousness (Blantyre 2), hypoglycaemia (2.1 mmol/L, corrected) and moderate anaemia (Hb 6.8 g/dL), RDT positive, 4% parasitaemia, no neck stiffness."

Problem list

1. Severe *P. falciparum* malaria, cerebral malaria to be confirmed after glucose and seizures are corrected.
2. Hypoglycaemia.
3. Convulsions.
4. Anaemia, not yet requiring transfusion.
5. Meningitis not yet excluded.

Model case presentation (ward round)

> "Baby J, a 2-year-old girl from Mukono, **weight 9.8 kg (weight-for-age Z −1.5), height 84 cm (height-for-age Z −0.9), MUAC 13.5 cm**, admitted 18 hours ago with 3 days of high fever, vomiting and refusal to breastfeed. She is the third child; she is **fully immunised for age by card** (BCG scar present; she missed the malaria vaccine because the rollout reached her district after her 8-month visit). She is **still breastfeeding** with family foods three times a day. Mother is **HIV negative** (tested in ANC); the child was not tested. They own a net but did not use it last week. On admission she was **prostrate**, temperature 39.4 °C, RR 52 with deep breathing, HR 168, CRT 2 s, **severe palmar pallor**, spleen 3 cm, Blantyre 4. RDT positive, slide 3+ (5% parasitaemia), **Hb 4.1 g/dL**, glucose 3.8 mmol/L. **Assessment: severe malaria with severe anaemia and respiratory distress.** She received IV artesunate 3 mg/kg (30 mg) at 0 and 12 h (third dose due at 24 h), whole blood 20 mL/kg over 4 h, and paracetamol. Today she is alert, breathing normally, feeding, Hb 7.6 g/dL post-transfusion. **Plan:** third artesunate dose at 24 h, then DP for 3 days; monthly DP for 3 months (PDMC); HIV test; iron and deworming at follow-up after the infection settles; Hb at day 14–28; net counselling; refer for malaria vaccine catch-up if her district policy allows."

EXAM FAVOURITE

Worked example 1: the artesunate dose. A 2-year-old weighs **11 kg** (under 20 kg) → **3 mg/kg × 11 = 33 mg**. A 60 mg vial reconstituted with 1 mL bicarbonate, then 5 mL saline = **10 mg/mL** → give **3.3 mL IV over about 5 minutes** (round to 3.4 mL is acceptable). For IM use, dilute with 2 mL saline = **20 mg/mL** → **1.65 mL IM into the anterior thigh**. A 7-year-old weighs **22 kg** (20 kg or more) → **2.4 mg/kg × 22 = 52.8 mg** ≈ 53 mg → **5.3 mL** of the 10 mg/mL solution. Doses at **0, 12 and 24 h**, then daily until the child can take oral medication; then a full 3-day ACT.

EXAM FAVOURITE

Worked example 2: hypoglycaemia and transfusion volumes. Same 11 kg child, glucose **1.8 mmol/L**: 10% dextrose **5 mL/kg = 55 mL** IV over 5–7 min. If only 50% is available: take **11 mL of 50% dextrose + 44 mL water for injection** (1 + 4 mL/kg) = 55 mL of 10%. Recheck in 15–30 min, then run a dextrose-containing maintenance fluid or start NGT feeds. Hb **3.6 g/dL** with deep breathing → transfuse **whole blood 20 mL/kg = 220 mL over 3–4 h** (or packed cells 10–15 mL/kg = 110–165 mL). Recheck Hb after the transfusion; repeat if Hb remains <4 g/dL or the child is still decompensated.

9. Management

Immediate / emergency

First 5 minutes

1. **Triage using ETAT**. Emergency signs: obstructed breathing, severe respiratory distress, central cyanosis, shock, **coma**, **convulsing**, severe dehydration. Any emergency sign → resuscitation room now.
2. **ABC**: open the airway, position (recovery position if unconscious), **oxygen** if SpO_2 <90% or severe respiratory distress, suction.
3. **Convulsing?** Diazepam 0.5 mg/kg rectally (or 0.2–0.3 mg/kg slow IV), max 10 mg.
4. **IV or IO access** and draw blood at once for: **glucose (bedside)**, **Hb/HemoCue**, RDT or blood slide, **group and cross-match**, blood culture if available.
5. **Glucose <2.5 mmol/L (or unknown in a comatose child) → 10% dextrose 5 mL/kg IV**.
6. **Shock?** Bolus only for true shock (complications table).
7. **Blantyre score** and weight. Then the **first dose of IV artesunate** within the first hour.

Severe malaria: the emergency algorithm

1. **Triage, ABC, oxygen**, recovery position if unconscious; stop seizures.
2. **IV access**; bedside **glucose, Hb, RDT**; cross-match; blood culture.
3. **Correct hypoglycaemia** (10% dextrose 5 mL/kg).
4. **Give IV artesunate** (IM into the anterior thigh if IV access fails) within the first hour. Weigh the child first.
5. **Shock?** Treat shock; otherwise **no fluid bolus**.
6. **Hb ≤4 g/dL, or 4–6 g/dL with distress, coma or shock** → transfuse.
7. **Coma or convulsions** → LP if safe; if not done, **add ceftriaxone 100 mg/kg/day** (meningitis dose). Very sick or anaemic child → antibiotics for possible bacteraemia.
8. **NGT** if unconscious (feeds, prevent aspiration and hypoglycaemia); catheter if needed for urine output.
9. **Monitor** (Monitoring and follow-up).
10. **Switch to oral ACT** after at least 3 artesunate doses (24 h) once the child can take oral drugs; give the **full 3-day course**.
11. **Discharge plan**: Hb check, post-discharge chemoprevention, nets, vaccine, follow-up.

> RED FLAG
> **Rectal artesunate is a bridge, not a cure.** In the CARAMAL project (Uganda, Nigeria and DRC), many children given rectal artesunate never completed referral or did not receive injectable treatment followed by an ACT, and outcomes were poor. Give it, **then make sure the child actually reaches a facility** that can give injectable artesunate and a full ACT.

Managing complications

Complication	Action (doses per kg)
Convulsions	**Diazepam 0.5 mg/kg rectally** or **0.2–0.3 mg/kg slow IV** (max 10 mg). Repeat once after 10 min if still fitting. Still fitting → **phenobarbital 15 mg/kg** IV over 15–20 min or IM (UCG: 10–15 mg/kg), then maintenance 2.5–5 mg/kg/day. Check glucose. **No prophylactic anticonvulsants.** Watch breathing after benzodiazepines plus phenobarbital
Hypoglycaemia (<2.5 mmol/L; UCG severe criterion <2.2)	**10% dextrose 5 mL/kg IV** over 5–7 min (1 mL/kg 50% + 4 mL/kg water). Recheck at 15–30 min. Then dextrose-containing maintenance fluid (e.g. 5–10% dextrose in half-strength Darrow's or RL as available) or NGT feeds. **Never undiluted 50% dextrose**
Severe anaemia	Transfuse if **Hb ≤4 g/dL**, or **Hb 4–6 g/dL with respiratory distress, impaired consciousness, shock, heart failure or hyperparasitaemia**. **Whole blood 20 mL/kg** or **packed cells 10–15 mL/kg** over **3–4 h**. In **SAM: 10 mL/kg whole blood once**. Furosemide **1 mg/kg IV** only if signs of fluid overload. Recheck Hb after
Shock (cold hands, CRT >3 s, weak fast pulse)	UCG: **0.9% saline 20 mL/kg** fast; WHO ETAT 2016: **10–20 mL/kg** then reassess. In SAM: 15 mL/kg over 1 h of RL with 5% dextrose (or half-strength Darrow's). Look for sepsis or bleeding; give antibiotics. **No boluses for children who are not in shock** (FEAST)
Acidosis (deep breathing)	Treat the cause: transfuse if anaemic, correct hypovolaemia only if shocked, oxygen. Bicarbonate is not routinely used in children
Hyperpyrexia	**Paracetamol 15 mg/kg** every 6 h; tepid sponging, undress
Coma	Recovery position, airway, NGT, turn 2-hourly, eye care, catheter; exclude hypoglycaemia (if no response within 20 min think of another cause), meningitis; look for subtle seizures
Renal impairment (urine <0.3 mL/kg/h after rehydration)	Rule out dehydration or shock; furosemide challenge **1 mg/kg** IV (WHO Pocket Book: 2 mg/kg, doubled hourly to max 8 mg/kg); refer for **peritoneal dialysis** if no response

Complication	Action (doses per kg)
Haemoglobinuria	Hydrate, check Hb and creatinine, transfuse if needed; continue artesunate
Bleeding	Fresh whole blood, vitamin K; avoid IM injections if possible
Possible bacteraemia / meningitis	**Ceftriaxone 100 mg/kg once daily** (max 4 g) IV/IM (or 50 mg/kg 12-hourly)

Definitive

Uncomplicated malaria (OPD, HC II upwards)

1. **Confirm** with RDT or slide.
2. **First-line: artemether–lumefantrine (AL)** for all ages (UCG 2023). Give the first dose in clinic and watch the child for 30 minutes.
3. **Paracetamol 10–15 mg/kg** every 6 hours for fever (max 4 doses in 24 h).
4. **Counsel**: give each dose with **milk or a fatty meal** (lumefantrine absorption needs fat), finish all 6 doses, sleep under a net, **return at once** with any danger sign (can't drink, vomits everything, convulsion, drowsy, fast breathing) or if fever persists after 2–3 days.
5. **If the child vomits within 30 minutes**, repeat the full dose; between 30 and 60 minutes, repeat half the dose. Vomiting repeatedly → treat as severe (parenteral).

Artemether–lumefantrine 20/120 mg tablets (UCG 2023)

Weight	Tablets per dose	Schedule	Total tablets
5–14 kg	**1**	0 h, 8 h, then twice daily on days 2 and 3 (24, 36, 48, 60 h)	6
15–24 kg	**2**	as above	12
25–34 kg	**3**	as above	18
≥35 kg	**4**	as above	24

Target dose: about **1.7 mg/kg artemether + 10 mg/kg lumefantrine** per dose. Dispersible tablets are available for young children.

> UGANDA CONTEXT
> **Infants under 5 kg.** UCG 2023 lists AL as first line for "all patients including children under 4 months", using the 5–14 kg band. The **latest WHO Guidelines for malaria** (consolidated version updated in September 2026) add a recommendation on a **new artemether–lumefantrine formulation for infants under 5 kg** (a lower-strength dispersible tablet). Check whether it is stocked in your facility before you prescribe; on most Ugandan wards a small infant still gets AL 20/120 dispersible 1 tablet per dose under senior supervision.

Severe malaria: completing treatment

- **Parenteral artesunate** at 0, 12 and 24 h, then daily until the child can take oral drugs (minimum 3 doses); then a **full 3-day oral ACT**. Alternatives only if artesunate is unavailable: IM artemether, then IV quinine (drug table).

UGANDA CONTEXT

Completing treatment and post-discharge chemoprevention (UCG 2023): discharge every child treated for severe malaria on **DHA–piperaquine (DP) for 3 days**, then review **monthly for 3 months** and give a monthly 3-day DP course (**post-discharge malaria chemoprevention, PDMC**). WHO recommends PDMC mainly for children admitted with **severe anaemia**; Uganda's guideline applies it to all severe malaria cases. Also **repeat the Hb before discharge and about 28 days later** (delayed haemolysis after artesunate).

Fluids and feeding

- **Maintenance (Holliday–Segar): 100 mL/kg/day for the first 10 kg + 50 mL/kg/day for the next 10 kg + 20 mL/kg/day for each kg above 20**. Example: 12 kg child = 1000 + 100 = **1100 mL/day (about 46 mL/h)**.
- Give **orally or by NGT** as soon as possible; IV fluids only while the child cannot feed. **Transfused blood counts towards the total.**
- The **most reliable sign of fluid overload** is an enlarging liver; others are basal crackles, gallop rhythm, eyelid oedema, raised JVP in older children.

Drug table

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Artemether–lumefantrine 20/120 mg (uncomplicated, first line)	By weight: **5–14 kg 1 tab; 15–24 kg 2; 25–34 kg 3; ≥35 kg (and adults) 4** (≈1.7 mg/kg artemether + 10 mg/kg lumefantrine per dose)	Oral, with milk or fatty food	0, 8, 24, 36, 48, 60 h	3 days (6 doses)	Repeat full dose if vomited within 30 min, half dose at 30–60 min
Artesunate + amodiaquine (alternative)	AS **4 mg/kg** + AQ **10 mg base/kg** (UCG age bands: 0–11 months ½ tab of each; 1–6 years 1 tab; 7–13 years 2 tabs); AS 50 mg tab + AQ 153 mg base tab	Oral	Once daily	3 days	First-line alternative
Dihydroartemisinin–piperaquine 40/320 mg (second line; completes severe malaria; PDMC)	DHA **4 mg/kg/day** + PPQ **18 mg/kg/day** (children under 25 kg need at least 2.5 mg/kg DHA and 20 mg/kg PPQ); UCG bands: 5–9.9 kg ½ tab; 10–20 kg 1; 20–40 kg 2	Oral	Once daily	3 days; PDMC monthly × 3 months	–
Pyronaridine–artesunate	UCG weight bands (granules 60/20 mg: 5–<8 kg 1 sachet; 8–<15 kg 2; 15–<20 kg 3; tablets 180/60 mg from 20 kg)	Oral	Once daily	3 days	Newer option
Quinine tablets (300 mg salt)	**10 mg/kg salt**	Oral	Every 8 h	7 days	Only if ACTs are unavailable; bitter, poor adherence

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
Artesunate IV/IM (severe, first line)	**3 mg/kg if <20 kg; 2.4 mg/kg if ≥20 kg** (adults 2.4 mg/kg)	IV over about 5 min (10 mg/mL) or IM anterior thigh (20 mg/mL)	0, 12 and 24 h, then **every 24 h**	Until able to take oral drugs (minimum 3 doses); then a full 3-day ACT	Reconstitute each vial with the 1 mL **5% sodium bicarbonate** supplied, then saline or 5% dextrose (never water for injection); use within 1 hour; delayed haemolysis: recheck Hb
Rectal artesunate (pre-referral, under 6 years only)	About 10 mg/kg: **100 mg** (1 supp) for 4 months to <3 years (5–<14 kg); **200 mg** (2 supps) for 3 to <6 years (14–19 kg)	Rectal	Once	Single pre-referral dose	Hold the buttocks together for 10 min; repeat if expelled within 30 min; **refer immediately**
Artemether (if no artesunate)	**3.2 mg/kg** loading, then **1.6 mg/kg**	IM (anterior thigh, never buttock)	Loading, then every 24 h	Until oral possible, then full ACT	Erratic absorption; second choice
Quinine dihydrochloride (if no artemisinin)	**20 mg/kg salt loading** (WHO Pocket Book; omit if quinine or mefloquine in the last 12–24 h), then **10 mg/kg**	IV infusion in 5% dextrose (10 mL/kg) over 2–4 h (loading over 4 h); **never as a bolus**	Every 8 h	Until oral possible, then ACT or quinine tablets to complete 7 days	Causes **hypoglycaemia**, tinnitus, arrhythmia; check glucose 2–4-hourly
10% dextrose	**5 mL/kg** (= 1 mL/kg 50% + 4 mL/kg water)	IV over 5–7 min	Recheck 15–30 min	Then maintenance dextrose or feeds	Never undiluted 50%
Diazepam	0.5 mg/kg PR or 0.2–0.3 mg/kg slow IV, **max 10 mg**	PR/IV	Repeat once after 10 min	–	Respiratory depression, especially with phenobarbital
Phenobarbital	**15 mg/kg** loading (UCG 10–15), then 2.5–5 mg/kg/day	IV over 15–20 min or IM	Load, then daily	–	No prophylactic use
Paracetamol	10–15 mg/kg (max 4 doses in 24 h); adult 1 g	Oral	6-hourly	While febrile	–
Ceftriaxone (possible meningitis or bacteraemia)	**100 mg/kg once daily** (max 4 g) or 50 mg/kg 12-hourly	IV/IM	Once daily	Per diagnosis	–
Whole blood / packed cells	**Whole blood 20 mL/kg** or **packed cells 10–15 mL/kg**; **SAM: 10 mL/kg whole blood**	IV	Over **3–4 h**	Repeat if Hb <4 g/dL or still decompensated	Fluid overload, transfusion reactions, transfusion-transmitted infection
Furosemide	**1 mg/kg** (WHO Pocket Book renal challenge: 2 mg/kg, doubled hourly to max 8 mg/kg)	IV	Once (overload or oliguria)	–	Only if overloaded

Drug	Dose (per kg + max for children; adult)	Route	Frequency	Duration	Key cautions
0.9% saline (true shock only)	**20 mL/kg** (UCG); 10–20 mL/kg then reassess (WHO ETAT 2016); SAM: 15 mL/kg over 1 h of RL with 5% dextrose	IV	Once, reassess	–	**No boluses without shock** (FEAST)

Referral criteria

Setting	What to do	Refer up when
VHT / HC II	RDT; AL for uncomplicated malaria; keep the child warm, breastfeeding or sugar water to prevent hypoglycaemia	**Danger signs → rectal artesunate (under 6) + first dose of antibiotic if indicated + refer**
HC III	As above plus **IM/IV artesunate** before referral, glucose check and dextrose, diazepam; admit stable cases (prostration only) if able to monitor	Coma, convulsions, severe anaemia, shock, renal impairment → HC IV or hospital
HC IV	Full severe malaria care: IV artesunate, dextrose, anticonvulsants, ceftriaxone, **transfusion if blood available**	Blood unavailable, persistent coma, renal failure → hospital or RRH
RRH / NRH (Mulago, Mengo, Lacor)	HDU care, LP, cultures, electrolytes, dialysis (peritoneal at selected sites), repeated transfusions, ICU	n/a

Monitoring and follow-up

- **Severe malaria**: Blantyre or GCS (on arrival and every 4 hours, more often if falling), vital signs, **glucose 4-hourly** (more often on quinine or in coma) and after every seizure, urine output, **liver size**, Hb daily.
- **Parasite clearance**: repeat the blood film at 24–48 h if not improving (artemisinin partial resistance).
- **After discharge**: monthly review for 3 months with PDMC; **Hb at about 7, 14 and 28 days** (delayed haemolysis); HIV test if not done; iron and deworming once the infection settles.
- **Uncomplicated malaria**: return if fever persists after 2–3 days or any danger sign; if fever returns within 3 weeks, do a **blood slide**.

10. Complications

Disease-related

System	Complications
Neurological	Cerebral malaria, status epilepticus, raised ICP and brain herniation, **neurological sequelae in about 10–20% of survivors of cerebral malaria** (hemiparesis, cortical blindness, ataxia, speech and behaviour problems), later **epilepsy** and learning difficulties
Haematological	Severe anaemia, thrombocytopenia, DIC, haemoglobinuria (blackwater)
Metabolic	Hypoglycaemia, lactic acidosis, hyponatraemia
Renal	Acute kidney injury (more common in older children)
Respiratory	Acidotic breathing, aspiration pneumonia (comatose child), pulmonary oedema (rare in children; fluid overload)
Infective	Bacteraemia (NTS), aspiration, hospital infections

System	Complications
Chronic	Iron deficiency, growth faltering, missed school; **hyperreactive malarial splenomegaly** in older children

Treatment-related

Complication	Cause	Recognition and action
Delayed haemolysis (PADH)	Artesunate (pitting)	Hb at 7, 14 and 28 days
Fluid overload, pulmonary oedema	Boluses or transfusion	Enlarging liver, crackles, gallop: furosemide 1 mg/kg; no boluses without shock
Hypoglycaemia, cinchonism (tinnitus), arrhythmia	Quinine	Glucose 2–4-hourly; never IV bolus
Transfusion reactions, transfusion-transmitted infection	Blood	Monitor during transfusion; transfuse only when indicated
Respiratory depression	Diazepam plus phenobarbital	Watch breathing; bag-valve-mask ready
Increased 48-hour mortality	Fluid boluses without shock (FEAST)	Maintenance only

11. Prognosis

- **AQUAMAT**: mortality of severe malaria in African children **8.5% with artesunate vs 10.9% with quinine**.
- **FEAST**: boluses in febrile African children with impaired perfusion (not hypotensive shock) increased 48-hour mortality (about **10.6% vs 7.3%**).
- **Cerebral malaria**: **neurological sequelae in about 10–20% of survivors** (hemiparesis, cortical blindness, ataxia, speech and behaviour problems), later epilepsy and learning difficulties.
- The **overlap** of cerebral malaria, severe anaemia and respiratory distress carries the highest mortality.
- Most deaths come from cerebral malaria, severe anaemia with acidosis, hypoglycaemia and **delays in referral**.

12. Prevention

Intervention	Details
Long-lasting insecticidal nets (LLINs)	Every child sleeps under a net every night; universal coverage campaigns plus ANC and EPI distribution. Newer dual-active-ingredient (e.g. chlorfenapyr) and PBO nets counter pyrethroid resistance
Indoor residual spraying (IRS)	Targeted districts (north and east)
Malaria vaccine	**Introduced into Uganda's routine schedule in April 2025** in 105 (later reported as 107) high-burden districts; **4 doses at 6, 7, 8 and 18 months**, 0.5 mL IM; the rollout will extend nationally. By January 2026, over 1 million children had received dose 1, 67% dose 2 and 47% dose 3 (MoH media brief)
Seasonal malaria chemoprevention (SMC)	**Karamoja** (seasonal transmission): monthly SP + amodiaquine for children 3–59 months during the rainy season
Post-discharge chemoprevention (PDMC)	Monthly DP for 3 months after severe malaria (UCG 2023)

Intervention	Details
Chemoprophylaxis for special groups	**Sickle cell disease** (UCG: SP; chloroquine 5 mg base/kg weekly as alternative; many clinics use other agents, check the SCD protocol); **HIV**: daily cotrimoxazole; non-immune visitors: mefloquine 5 mg/kg weekly
IPTp for the mother	SP monthly from 13 weeks (not with cotrimoxazole) protects the newborn's birth weight
Environment	Drain stagnant water, larviciding, screens, repellents in the evening

The two vaccines (both WHO-recommended)

	RTS,S/AS01 (Mosquirix)	R21/Matrix-M
WHO recommendation	2021	October 2023
Target	Circumsporozoite protein (pre-erythrocytic)	Same antigen, more antigen per dose, cheaper, larger supply
Efficacy	About **36%** against clinical malaria over 4 years with a booster (5–17 months at first dose, phase 3, Lancet 2015)	About **75%** in seasonal sites and **68%** in standard (perennial) sites over 12 months after 3 doses (phase 3, Lancet 2024)
Schedule	4 doses from 5 months	3 doses one month apart + a 4th dose about 12 months later

UGANDA CONTEXT

Uganda's schedule is **6, 7, 8 and 18 months**. The vaccine **does not replace nets, testing or ACTs**: a vaccinated child with fever still needs a malaria test the same day. WHO has been reviewing **3-dose versus 4-dose schedules in perennial transmission** (September 2026 guideline update); follow the UNEPI schedule your district uses.

13. Special Populations

- **Infants under 6 months**: partly protected by maternal antibody and fetal haemoglobin, but malaria is a red flag; **under 5 kg**: see the WHO 2026 infant AL formulation (Uganda callout in section 9).
- **Severe acute malnutrition**: hypoglycaemia threshold **<3 mmol/L**; **transfuse 10 mL/kg whole blood once**; shock fluid 15 mL/kg over 1 h of RL with 5% dextrose (or half-strength Darrow's); no standard boluses.
- **Sickle cell disease**: high mortality with malaria; sequestration and severe anaemia; chemoprophylaxis per SCD protocol.
- **HIV**: more severe malaria; daily cotrimoxazole is protective; IPTp SP not with cotrimoxazole in mothers.
- **Non-immune and older children** (highlands, travellers): adult-type disease with cerebral malaria, renal failure and jaundice; artesunate 2.4 mg/kg from 20 kg.
- **Children under 6 years at HC II or community level**: rectal artesunate pre-referral only.

14. Key Points

High-yield points

- **Test before you treat**, but never delay artesunate in a child with danger signs.
- **First 5 minutes**: Blantyre, glucose, Hb; then artesunate.

- AL by weight band, 6 doses with fatty food.
- IV artesunate **3 mg/kg under 20 kg**, 2.4 mg/kg from 20 kg, at 0, 12, 24 h then daily, then a full ACT.
- Hypoglycaemia: **10% dextrose 5 mL/kg**; never undiluted 50%.
- Pale with deep breathing = anaemia with acidosis: **give blood**, not a bolus.
- **No fluid boluses** unless truly shocked (FEAST).
- Coma + positive RDT is not cerebral malaria until glucose, seizures and meningitis are dealt with.
- Very sick child with malaria: add antibiotics (NTS bacteraemia).
- Discharge after severe malaria: DP, PDMC for 3 months, Hb recheck, net and vaccine.

Numbers & doses

Item	Value
AL 20/120 mg	**5–14 kg 1 tab; 15–24 kg 2; 25–34 kg 3; ≥35 kg 4**; at **0, 8, 24, 36, 48, 60 h**, with fatty food
ASAQ	AS 4 mg/kg + AQ 10 mg/kg once daily × 3 days
DP	DHA 4 mg/kg + PPQ 18 mg/kg once daily × 3 days
IV/IM artesunate	**3 mg/kg (<20 kg); 2.4 mg/kg (≥20 kg)** at 0, 12, 24 h then daily; IV 10 mg/mL, IM 20 mg/mL
Rectal artesunate	**100 mg** (4 months–<3 years); **200 mg** (3–<6 years); single dose, then refer
IM artemether	3.2 mg/kg loading, then 1.6 mg/kg daily
IV quinine	20 mg/kg loading over 4 h, then 10 mg/kg 8-hourly over 2–4 h
Hypoglycaemia threshold	**<2.5 mmol/L** (WHO Pocket Book; <3 in SAM); UCG severe criterion <2.2
10% dextrose	**5 mL/kg** IV (= 1 mL/kg 50% + 4 mL/kg water)
Severe anaemia (UCG)	Hb **<5 g/dL**
Transfuse	Hb ≤4, or 4–6 with complications; **whole blood 20 mL/kg or packed cells 10–15 mL/kg over 3–4 h**
Diazepam	0.5 mg/kg PR or 0.2–0.3 mg/kg IV (max 10 mg); repeat once
Phenobarbital	15 mg/kg loading (UCG 10–15)
Paracetamol	10–15 mg/kg 6-hourly
Ceftriaxone (meningitis cover)	100 mg/kg once daily (max 4 g)
Blantyre scale	Max 5; **≤2 = coma**
Repeated convulsions	**≥2 in 24 h**
Hyperparasitaemia	**>10%**
RDT stays positive	**2–3 weeks** after cure (HRP2)
Maintenance fluids	100/50/20 mL/kg/day
Malaria vaccine (Uganda)	**6, 7, 8, 18 months** (since April 2025)
AQUAMAT mortality	8.5% artesunate vs 10.9% quinine
Lower limit systolic BP (1–10 y)	**70 + (2 × age)** mmHg

Exam traps

- **"RDT positive, so the coma is cerebral malaria."** Wrong: correct glucose, stop seizures, wait 1 hour, exclude meningitis.

- **"Give 2.4 mg/kg artesunate to every child."** Wrong: children **under 20 kg need 3 mg/kg**.
- **"The child is pale and breathing fast: give a 20 mL/kg saline bolus."** Wrong: this is acidosis from anaemia; **give blood**. Boluses only for true shock (FEAST).
- **"Stop after the IV doses once the child wakes up."** Wrong: always complete a **full 3-day oral ACT**.
- **"Fever 10 days after AL and the RDT is positive, so it's treatment failure."** Not necessarily: the **HRP2 RDT stays positive 2–3 weeks**; do a **blood slide**.
- **"Rectal artesunate treats severe malaria at the HC II."** Wrong: it is **pre-referral only** and only for children under 6.
- **"Give 50% dextrose IV push."** Wrong in children: dilute to **10%, 5 mL/kg**.
- **"The vaccine makes nets unnecessary."** Wrong: efficacy is partial; use every tool.

Questions seniors ask

Viva questions

1. *Name five features of severe malaria in children.* Impaired consciousness or coma, prostration, ≥2 convulsions in 24 h, respiratory distress or acidosis, Hb <5 g/dL, glucose <2.2 mmol/L, shock, haemoglobinuria, jaundice, bleeding.
2. *Dose of IV artesunate?* 3 mg/kg under 20 kg; 2.4 mg/kg from 20 kg; at 0, 12, 24 h then daily; then full oral ACT.
3. *Why is artesunate preferred to quinine?* AQUAMAT: lower mortality (8.5% vs 10.9%), less hypoglycaemia, easier to give.
4. *What is the Blantyre coma scale?* Eyes (0–1), motor (0–2), verbal (0–2); ≤2 = coma.
5. *How do you treat hypoglycaemia?* 10% dextrose 5 mL/kg IV, recheck in 15–30 min, then maintenance dextrose or feeds.
6. *Which malaria vaccine schedule does Uganda use?* 4 doses at 6, 7, 8 and 18 months (introduced April 2025).
7. *Why is a fluid bolus dangerous in severe malaria?* FEAST: boluses in febrile African children with impaired perfusion (not hypotensive shock) **increased** 48-hour mortality (about 10.6% vs 7.3%).

On the ward round

- **What do you check first in a febrile child who is drowsy?** Airway and breathing, then glucose, Hb and the Blantyre score, while getting IV access and giving the first dose of artesunate.
- **This 15 kg child needs IV artesunate. What dose and how do you prepare it?** 3 mg/kg × 15 = 45 mg. Reconstitute a 60 mg vial with 1 mL 5% sodium bicarbonate, add 5 mL saline (10 mg/mL), give 4.5 mL IV over about 5 minutes; repeat at 12 and 24 h, then daily.
- **When will you transfuse?** Hb ≤4 g/dL, or 4–6 g/dL with respiratory distress, shock, impaired consciousness, heart failure or hyperparasitaemia: whole blood 20 mL/kg (or packed cells 10–15 mL/kg) over 3–4 h; 10 mL/kg in SAM.
- **Why don't you give a bolus to this breathless pale child?** The breathlessness is metabolic acidosis from anaemia, not hypovolaemia; FEAST showed boluses increased death in febrile African children with poor perfusion. Blood is the treatment.
- **The coma persists after glucose and diazepam. What next?** Recheck glucose, look for subtle seizures, examine for raised ICP; LP if safe, otherwise give ceftriaxone; continue artesunate; nurse in the recovery position with an NGT.
- **What will you do at discharge?** Complete DP, arrange monthly DP for 3 months (PDMC), check Hb at about 2–4 weeks for delayed haemolysis, HIV test if not done, iron and deworming once well, net and vaccine counselling, and explain danger signs.
- **Why does the child need antibiotics if the slide is positive?** Parasitaemia is common in well children in Uganda and bacteraemia (especially NTS) often coexists with severe malaria and anaemia; meningitis cannot be excluded without an LP.

Mnemonics

- **Severe malaria in children: "CAPS HD"**: **C**onvulsions or Coma, **A**naemia (Hb <5) and **A**cidosis, **P**rostration, **S**hock, **H**ypoglycaemia and **H**aemoglobinuria, **D**istress (respiratory) and **D**ark urine or renal.
- **Artesunate "3 under 20"**: 3 mg/kg under 20 kg; 2.4 mg/kg from 20 kg; doses at **0-12-24** then daily.
- **AL bands "5-15-25-35 → 1-2-3-4"**: weight 5, 15, 25, 35 kg → 1, 2, 3, 4 tablets per dose.
- **Blantyre "1-2-2"**: eyes 1, motor 2, verbal 2 (max 5); "**2 or less, coma: assess for the rest**" (glucose, seizures, meningitis).
- **Dextrose "10 × 5"**: 10% dextrose, 5 mL/kg.
- **Transfusion "20 or 10"**: whole blood 20 mL/kg, packed cells 10 mL/kg (10 mL/kg whole blood if SAM).

OSCE tips

- **ETAT station**: say "I will check the glucose" **before** anyone asks.
- **Blantyre station**: show the Blantyre score with a real stimulus (nail-bed pressure with a pen).
- **Dosing**: calculate doses aloud with the weight.
- **Counselling station**: explain AL with food, the 6-dose schedule using the blister pack, and the danger signs for return.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	First-line treatment for uncomplicated malaria in a Ugandan child (UCG 2023)?	**Artemether–lumefantrine (AL) 20/120 mg**, 6 doses at 0, 8, 24, 36, 48 and 60 h, with milk or fatty food. Alternative: artesunate–amodiaquine; second line: DHA–piperaquine.
2	AL weight bands (tablets per dose)?	**5–14 kg: 1 · 15–24 kg: 2 · 25–34 kg: 3 · ≥35 kg: 4** ("5-15-25-35 → 1-2-3-4").
3	IV artesunate dose in children?	**3 mg/kg if under 20 kg; 2.4 mg/kg if 20 kg or more**, at 0, 12 and 24 h, then every 24 h until oral treatment is possible; then a **full 3-day ACT**.
4	How is IV artesunate prepared?	Reconstitute the vial with the **1 mL 5% sodium bicarbonate** supplied; for IV add **5 mL saline or 5% dextrose (10 mg/mL)** and give over about 5 min; for IM add **2 mL saline (20 mg/mL)** into the anterior thigh. Never water for injection. Use within 1 hour.
5	Pre-referral rectal artesunate doses (UCG 2023)?	Children **under 6 years only**: **100 mg** (1 suppository) for 4 months to <3 years (5–<14 kg); **200 mg** (2 suppositories) for 3 to <6 years (14–19 kg). Hold buttocks 10 min; repeat if expelled within 30 min; refer immediately.
6	Blantyre coma scale components?	**Eyes**: watches/follows 1, fails 0. **Motor**: localises pain 2, withdraws 1, nil/non-specific 0. **Verbal**: appropriate cry 2, moan/abnormal cry 1, none 0. Max 5; **≤2 = unrousable coma**.
7	Definition of cerebral malaria?	Unrousable coma (Blantyre ≤2) with *P. falciparum* parasitaemia, persisting **more than 1 hour after a seizure**, with **hypoglycaemia corrected** and other causes (meningitis, encephalitis) excluded.
8	Severe malarial anaemia (UCG 2023)?	**Hb <5 g/dL** (PCV <15%) with parasitaemia.
9	Transfusion criteria and volume in severe malaria?	Transfuse if **Hb ≤4 g/dL**, or **4–6 g/dL with respiratory distress, impaired consciousness, shock, heart failure or hyperparasitaemia**. **Whole blood 20 mL/kg** or **packed cells 10–15 mL/kg** over 3–4 h (10 mL/kg whole blood once in SAM).
10	Treatment of hypoglycaemia in a child with malaria?	**10% dextrose 5 mL/kg IV** over 5–7 min (make it from 1 mL/kg of 50% + 4 mL/kg water). Recheck in 15–30 min; then dextrose-containing maintenance or NGT feeds. Never undiluted 50%.
11	Hypoglycaemia threshold in severe malaria?	UCG 2023 severe criterion: **<2.2 mmol/L** (40 mg/dL). WHO Pocket Book treats below **2.5 mmol/L** (**<3 mmol/L in severe malnutrition**).
12	Anticonvulsant doses in severe malaria?	**Diazepam 0.5 mg/kg rectally** or **0.2–0.3 mg/kg slow IV** (max 10 mg), repeat once after 10 min; then **phenobarbital 15 mg/kg** IV/IM loading (UCG 10–15 mg/kg). Always check glucose.
13	What did the AQUAMAT trial show?	In 5,425 African children with severe malaria, **artesunate reduced mortality to 8.5% vs 10.9% with quinine** (22.5% relative reduction), with fewer convulsions, less coma and less hypoglycaemia (Lancet 2010).
14	What did the FEAST trial show?	Fluid boluses in febrile African children with impaired perfusion (most had malaria or sepsis) **increased 48-h mortality** (about 10.6% vs 7.3%). Give boluses only for true shock.
15	How long does an HRP2 RDT stay positive after cure?	About **2–3 weeks** (sometimes longer). If fever returns within 3 weeks of treatment, use a **blood slide**.

#	Question	Answer
16	When does UCG 2023 prefer a blood slide to an RDT?	Symptoms persist after completing treatment; child returns within 3 weeks of treatment; RDT negative with no other cause for fever. The thin film also gives **% parasitaemia** to follow response.
17	What is post-discharge malaria chemoprevention (PDMC) in Uganda?	UCG 2023: discharge every severe malaria case on **DHA–piperaquine**, then a **monthly 3-day DP course for 3 months**, with monthly review. WHO targets children admitted with severe anaemia.
18	Why recheck Hb 1–4 weeks after severe malaria treated with artesunate?	**Post-artesunate delayed haemolysis (PADH)**: red cells 'pitted' of parasites by the spleen are destroyed 1–3 weeks later. UCG: repeat Hb before discharge and about 28 days later.
19	Uganda's malaria vaccine schedule?	Introduced into routine immunisation in **April 2025** in 107 high-burden districts: **4 doses at 6, 7, 8 and 18 months** (0.5 mL IM).
20	R21/Matrix-M phase 3 efficacy?	About **75% in seasonal** and **68% in standard (perennial) transmission sites** over 12 months after 3 doses (Lancet 2024); a booster a year later maintained protection. WHO recommended it in October 2023.
21	Why does falciparum malaria cause coma when the peripheral count may be low?	**Sequestration**: mature parasites express PfEMP1 and stick to endothelium in the brain and other organs, so they are hidden from the peripheral film while causing microvascular obstruction, inflammation and brain swelling.
22	Respiratory distress in severe malaria: commonest mechanism?	**Metabolic (lactic) acidosis**, usually with severe anaemia, not pneumonia. Deep (Kussmaul) breathing settles with **blood transfusion** and correction of perfusion.
23	Quinine IV regimen if artemisinins unavailable?	**20 mg/kg salt loading** in 10 mL/kg 5% dextrose over 2–4 h (omit if quinine in last 12–24 h), then **10 mg/kg every 8 h** over 2–4 h; never as a bolus. Watch for **hypoglycaemia**.
24	Maintenance fluids for a 12 kg child?	Holliday–Segar 100/50/20: 10 × 100 + 2 × 50 = **1100 mL/day ≈ 46 mL/h**. Give orally/NGT when possible; transfused blood counts.
25	Artemisinin partial resistance in Uganda?	**kelch13 mutations** (e.g. A675V, C469Y) with delayed parasite clearance were documented in **northern Uganda** (NEJM 2021). Partner drugs still cure most; complete the full course and check clearance if the child does not improve.
26	Most reliable sign of fluid overload in a child?	An **enlarging liver** (also basal crackles, gallop, eyelid oedema). Measure the liver edge on admission as a baseline.
27	Malaria prevention package in Uganda?	LLINs, IRS, **malaria vaccine** (6, 7, 8, 18 months), **SMC in Karamoja** (SP + amodiaquine monthly in the rainy season, 3–59 months), PDMC after severe malaria, IPTp with SP for mothers, cotrimoxazole in HIV, environmental control.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old boy weighing 11 kg is brought to Mengo emergency unit after two generalised convulsions at home. He is unresponsive: he does not follow his mother's face, withdraws his arm to nail-bed pressure and moans. Temperature 39.2 °C, RR 44, HR 160, CRT 2 s. Malaria RDT is positive. Bedside glucose is 1.9 mmol/L. What is the MOST appropriate immediate action?

- A Give IV artesunate and observe the conscious level for 1 hour
- B Give 50% dextrose 1 mL/kg as an IV push
- C Give 10% dextrose 55 mL IV over 5–7 minutes and recheck glucose in 15–30 minutes
- D Insert a nasogastric tube and give sugar water before any IV drug
- E Give 0.9% saline 20 mL/kg as a bolus to improve cerebral perfusion

2. A 4-year-old girl weighing 16 kg is admitted to a HC IV with severe malaria (prostration, RDT positive, Hb 7.2 g/dL, glucose 4.8 mmol/L). She is alert enough to cry but cannot sit. What is the correct first dose of IV artesunate?

- A 48 mg (3 mg/kg)
- B 38.4 mg (2.4 mg/kg)
- C 160 mg (10 mg/kg)
- D 51 mg (3.2 mg/kg)
- E 25.6 mg (1.6 mg/kg)

3. An 18-month-old boy weighing 9 kg from Kasese lowlands has had fever for 4 days. He is very pale with severe palmar pallor, RR 64 with deep breathing and no chest indrawing, HR 170, CRT 2 s, chest clear, liver 2 cm. Blantyre score 5. RDT positive. HemoCue Hb 3.5 g/dL. Glucose 5.1 mmol/L. SpO2 95%. Which management step will MOST improve his breathing?

- A Oxygen by nasal prongs at 1 L/min
- B 0.9% saline 20 mL/kg over 15 minutes
- C IV ceftriaxone for presumed pneumonia
- D Furosemide 1 mg/kg IV
- E Whole blood 180 mL over 3–4 hours

4. A 3-year-old girl (13 kg) at a HC II in Kayunga has had fever for 2 days. She now vomits everything and cannot sit up. The RDT is positive. The nearest HC IV is 2 hours away and the HC II has no injectable antimalarials. What should the health worker do?

- A Give the first dose of AL and repeat if she vomits within 30 minutes
- B Give rectal artesunate 100 mg (one suppository) and refer immediately
- C Give rectal artesunate 200 mg (two suppositories) and refer immediately
- D Give rectal artesunate and keep her at the HC II for daily doses
- E Give paracetamol and review in 24 hours

5. A 5-year-old boy (18 kg) admitted with cerebral malaria has had three convulsions in the ward despite diazepam 0.5 mg/kg rectally given twice 10 minutes apart. Glucose is 4.6 mmol/L, SpO2 96%, airway protected. What is the MOST appropriate next drug?

- A A third dose of rectal diazepam
- B Oral carbamazepine via nasogastric tube
- C Paracetamol 15 mg/kg and tepid sponging only
- D Phenobarbital 270 mg IV over 15–20 minutes
- E IV dexamethasone 0.15 mg/kg for cerebral oedema

6. A 6-year-old girl (19 kg) completed a supervised 3-day course of AL 11 days ago for slide-confirmed malaria and recovered. She now presents with 1 day of fever 38.6 °C, no danger signs. An HRP2-based RDT is positive. What is the BEST next step?

- A Request a thick and thin blood slide before deciding on antimalarial treatment
- B Diagnose AL treatment failure and give DHA–piperaquine
- C Repeat AL because the RDT is positive
- D Admit and give IV artesunate
- E Reassure: a positive RDT after treatment is always a false positive

7. A 2-year-old (10 kg) with severe malaria and Blantyre score 2 has received IV artesunate and dextrose. One hour later glucose is 5.2 mmol/L, there has been no further seizure, and he remains Blantyre 2. There is no papilloedema, pupils are equal and reactive, he is haemodynamically stable and the platelet count is 110 x 10^9/L. What should you do next?

- A Give a further dose of artesunate now
- B Start IV mannitol
- C Perform a lumbar puncture
- D Give phenobarbital prophylaxis
- E Wait 24 hours and repeat the RDT

8. A 9-month-old girl (7.5 kg, MUAC 12.8 cm) is seen at OPD with 2 days of fever. She is breastfeeding well, alert, RR 42, no vomiting. RDT positive. Which prescription is correct?

- A AL 20/120 mg, 2 tablets per dose for 6 doses
- B AL 20/120 mg, 1 tablet once daily for 3 days
- C IM artesunate 3 mg/kg, then oral AL
- D Quinine tablets 10 mg/kg every 8 hours for 7 days
- E AL 20/120 mg, 1 tablet per dose at 0, 8, 24, 36, 48 and 60 hours with breast milk or food

9. A 4-year-old boy (15 kg) with severe malaria is in the ward. Since admission he has received 0.9% saline at 100 mL/h for 6 hours plus 300 mL of whole blood. He now has RR 58, fine basal crackles, a gallop rhythm and the liver edge has moved from 2 cm to 5 cm below the costal margin. What is the MOST appropriate action?

- A Give another 20 mL/kg bolus because he is tachycardic
- B Stop IV fluids, sit him up, give oxygen and furosemide 1 mg/kg IV
- C Start a second unit of blood
- D Change to 10% dextrose at the same rate
- E Give IV ceftriaxone for hospital-acquired pneumonia and continue fluids

10. A 7-year-old girl (24 kg) is ready for discharge after 3 days of IV artesunate for severe malaria with severe anaemia (transfused once). She is eating well, Hb 8.4 g/dL. Which discharge plan follows UCG 2023?

A Stop antimalarials because IV artesunate has cleared the parasites
B AL for 3 days, then review only if fever recurs
C Weekly chloroquine prophylaxis until the rainy season ends
D DHA–piperaquine for 3 days, then monthly 3-day DP courses for 3 months with Hb rechecked at about 28 days
E Primaquine 0.25 mg/kg single dose and discharge

11. Which statement about the malaria vaccine in Uganda is correct?

A It is given as a single dose at birth with BCG
B It replaces the need for bed nets in vaccinated children
C It is given as 4 doses at 6, 7, 8 and 18 months in the routine schedule introduced in April 2025
D It is contraindicated in HIV-exposed infants
E A vaccinated child with fever can be treated without a malaria test

12. A 3-year-old boy (14 kg) is admitted with severe malaria. On day 2, the nurse reports passing of dark 'Coca-Cola' urine. Dipstick shows 3+ blood but microscopy shows no red cells. Urine output is 1.2 mL/kg/h. Hb has fallen from 7.8 to 6.1 g/dL. He is alert and breathing normally. What is the BEST interpretation and plan?

A Haemoglobinuria from intravascular haemolysis: continue artesunate, maintain hydration, monitor Hb and renal function, transfuse if criteria are met
B Acute glomerulonephritis: restrict fluids and give furosemide
C Urinary tract infection: start ciprofloxacin
D Schistosoma haematobium: give praziquantel
E Artesunate allergy: switch to IV quinine

13. A 4-year-old girl (16 kg) arrives with severe malaria. She has cold hands, CRT 4 seconds, a weak fast pulse of 180/min, and is lethargic. Hb is 9.1 g/dL, glucose 4.0 mmol/L. She is not malnourished. What fluid should you give first?

A No fluid; give maintenance via NGT only
B 5% dextrose 20 mL/kg over 15 minutes
C Whole blood 20 mL/kg over 4 hours
D Half-strength Darrow's 15 mL/kg over 1 hour
E 0.9% saline (or Ringer's lactate) 10–20 mL/kg bolus, then reassess

14. A 5-year-old (18 kg) is being treated with IV quinine because artesunate is out of stock. Six hours after the loading dose he becomes restless and sweaty, then drowsy. Blantyre was 5 on admission, now 3. What is the FIRST thing to check?

A Serum sodium
B Bedside blood glucose
C Repeat blood slide
D Lumbar puncture
E ECG for QT prolongation

15. Which feature on its own classifies a parasitaemic child as having SEVERE malaria under UCG 2023?

- A Temperature 38.9 °C
- B Two vomits in 24 hours but able to drink
- C Palpable spleen 4 cm below the costal margin
- D Two generalised convulsions in the past 24 hours
- E Hb 7.5 g/dL

Answers & explanations

1. C · Give 10% dextrose 55 mL IV over 5–7 minutes and recheck glucose in 15–30 minutes

C is correct: his Blantyre score is 2 (eyes 0, motor 1, verbal 1) and glucose is below 2.5 mmol/L, so hypoglycaemia must be corrected at once with **10% dextrose 5 mL/kg (11 kg × 5 = 55 mL)**, then rechecked; artesunate follows within the hour. A is wrong: artesunate is essential but leaving hypoglycaemia untreated can cause permanent brain injury, and the coma may be due to the low glucose. B is wrong: undiluted 50% dextrose is hyperosmolar and damages veins; UCG says never give it to children. D is wrong: NGT sugar is a fallback only when IV access is impossible. E is wrong: he is not in shock (CRT 2 s) and FEAST showed boluses increase mortality in such children.

2. A · 48 mg (3 mg/kg)

A is correct: children **under 20 kg** need **3 mg/kg** (WHO since 2015; UCG 2023) because they have lower drug exposure at 2.4 mg/kg; 16 × 3 = 48 mg, at 0, 12 and 24 h. B is wrong: 2.4 mg/kg is the dose for children of 20 kg or more. C is wrong: 10 mg/kg is the rectal artesunate pre-referral dose. D is wrong: 3.2 mg/kg is the IM artemether loading dose. E is wrong: 1.6 mg/kg is the daily IM artemether maintenance dose.

3. E · Whole blood 180 mL over 3–4 hours

E is correct: deep breathing with very severe anaemia and a clear chest is **metabolic acidosis from anaemia**; the treatment is transfusion: whole blood **20 mL/kg (9 × 20 = 180 mL) over 3–4 h** (Hb ≤4 g/dL is an absolute indication). A is wrong: his SpO2 is normal; oxygen does not fix the lack of haemoglobin. B is wrong: he is not shocked, and FEAST showed boluses increase death; it would also dilute his Hb further. C is wrong: chest is clear with no indrawing; antibiotics may be considered for bacteraemia but will not settle the acidotic breathing. D is wrong: there are no signs of fluid overload; furosemide is only for overload during transfusion.

4. B · Give rectal artesunate 100 mg (one suppository) and refer immediately

B is correct: vomiting everything and prostration are danger signs (severe malaria). At a facility without injectables, UCG 2023 gives **pre-referral rectal artesunate** (children under 6) and **immediate referral**; at 13 kg (5–<14 kg band, under 3 years) the dose is **100 mg = one suppository**. A is wrong: oral treatment is inappropriate when she vomits everything and is prostrate. C is wrong: 200 mg is for children of 3 to <6 years weighing 14–19 kg. D is wrong: rectal artesunate is a bridge only; CARAMAL showed poor outcomes when referral and injectable treatment plus ACT were not completed. E is wrong: delay can be fatal within hours.

5. D · Phenobarbital 270 mg IV over 15–20 minutes

D is correct: after two benzodiazepine doses, give a second-line drug: **phenobarbital 15 mg/kg** (18 × 15 = 270 mg) IV slowly (or IM), monitoring breathing. A is wrong: more than two benzodiazepine doses adds respiratory depression without much benefit. B is wrong: carbamazepine is a maintenance drug with no role in acute status. C is wrong: fever control helps but will not stop ongoing seizures. E is wrong: steroids are harmful in cerebral malaria (older trials showed prolonged coma) and are not recommended.

6. A · Request a thick and thin blood slide before deciding on antimalarial treatment

A is correct: HRP2 antigen persists for **2–3 weeks after cure**, so a positive RDT within 3 weeks does not prove new or persistent infection; UCG 2023 recommends a **blood slide** in this situation, plus a search for other causes of fever. B is wrong: failure must be confirmed by microscopy before changing drugs. C is wrong: retreating on an RDT alone may miss the real cause (UTI, pneumonia, typhoid). D is wrong: she has no severe features. E is wrong: she may have a genuine new infection; 'always' is untrue.

7. C · Perform a lumbar puncture

C is correct: persistent coma after correction of glucose and seizures needs **meningitis excluded**; with no signs of raised ICP, stable circulation and adequate platelets, **LP is safe** (if not possible, give ceftriaxone). A is wrong: the next artesunate dose is due at 12 h; extra doses do not speed recovery. B is wrong: mannitol has not been shown to help cerebral malaria and is not routine. D is wrong: prophylactic anticonvulsants are not recommended (phenobarbital prophylaxis increased mortality when given without ventilatory support). E is wrong: the RDT is already positive; repeating it adds nothing.

8. E · AL 20/120 mg, 1 tablet per dose at 0, 8, 24, 36, 48 and 60 hours with breast milk or food

E is correct: uncomplicated malaria in a 5–14 kg child = **1 tablet AL per dose, 6 doses over 3 days**, taken with milk or a fatty meal for lumefantrine absorption. A is wrong: 2 tablets is the 15–24 kg band. B is wrong: AL is given twice daily (6 doses), not once daily. C is wrong: she has no severe features, so parenteral treatment is not indicated. D is wrong: quinine is reserved for when ACTs are unavailable; adherence to 21 doses is poor.

9. B · Stop IV fluids, sit him up, give oxygen and furosemide 1 mg/kg IV

B is correct: an enlarging liver, gallop and crackles mean **fluid overload**; he received far more than maintenance (15 kg ≈ 1250 mL/day ≈ 52 mL/h) plus 20 mL/kg blood. Stop IV fluids, sit up, oxygen, **furosemide 1 mg/kg** (UCG 1–2 mg/kg). A is wrong: more fluid worsens pulmonary oedema. C is wrong: no indication for further blood now, and it would add volume. D is wrong: changing the fluid type does not reduce volume. E is wrong: the picture is overload, not infection; continuing fluids is harmful.

10. D · DHA–piperaquine for 3 days, then monthly 3-day DP courses for 3 months with Hb rechecked at about 28 days

D is correct: UCG 2023 says discharge all severe malaria cases on **DP**, review monthly for 3 months and give **monthly DP (post-discharge malaria chemoprevention)**, and repeat the Hb around **28 days** because of delayed post-artesunate haemolysis and re-infection. A is wrong: parenteral artesunate must always be followed by a full oral ACT. B is wrong: AL completes treatment but misses PDMC and the Hb follow-up. C is wrong: chloroquine is not used for prevention in this group (high resistance). E is wrong: single low-dose primaquine is a transmission-blocking measure, not treatment completion.

11. C · It is given as 4 doses at 6, 7, 8 and 18 months in the routine schedule introduced in April 2025

C is correct: the MoH introduced the malaria vaccine into routine immunisation in **April 2025** in 107 high-burden districts, **4 doses at 6, 7, 8 and 18 months**. A is wrong: it is not given at birth. B is wrong: efficacy is partial (RTS,S about 36% over 4 years; R21 about 68–75% over the first year), so nets and prompt testing remain essential. D is wrong: HIV exposure is not a contraindication. E is wrong: every febrile child still needs a malaria test (UCG 'test before treat').

12. A · Haemoglobinuria from intravascular haemolysis: continue artesunate, maintain hydration, monitor Hb and renal function, transfuse if criteria are met

A is correct: dipstick blood without red cells on microscopy = **free haemoglobin (haemoglobinuria)** from intravascular haemolysis, a severe malaria feature; continue artesunate, keep him hydrated, watch creatinine and urine output, and transfuse if Hb falls to ≤4 or he decompensates. B is wrong: AGN gives red cells/casts, oedema and hypertension. C is wrong: there is no pyuria or symptoms of UTI. D is wrong: schistosomal haematuria contains red cells and is chronic. E is wrong: haemolysis is part of the disease (or delayed post-artesunate haemolysis), not an allergy; quinine is less effective.

13. E · 0.9% saline (or Ringer's lactate) 10–20 mL/kg bolus, then reassess

E is correct: cold hands + CRT >3 s + weak fast pulse = **shock** (ETAT), so a crystalloid bolus is indicated: UCG 2023 gives 20 mL/kg 0.9% saline or Ringer's lactate as rapidly as possible; WHO ETAT 2016 says 10–20 mL/kg over 30–60 min and reassess after each bolus; then look for sepsis and give antibiotics. A is wrong: FEAST argues against boluses for children *without* shock, but this child has established shock. B is wrong: dextrose solutions do not stay in the circulation and cause hyperglycaemia. C is wrong: her Hb is 9.1 g/dL; blood is for severe anaemia. D is wrong: the slower Darrow's/RL-dextrose regimen is for children with severe acute malnutrition.

14. B · Bedside blood glucose

B is correct: quinine stimulates **insulin release** and causes **hypoglycaemia**, which presents with sweating, restlessness and falling consciousness; glucose takes seconds and treatment (10% dextrose 5 mL/kg) is immediate. A is wrong: hyponatraemia is possible but slower to check and less likely. C is wrong: parasite count will not explain an acute fall over minutes. D is wrong: LP may be needed later, but only after glucose is checked and corrected. E is wrong: QT prolongation causes arrhythmia, not this picture, and an ECG comes after the glucose.

15. D · Two generalised convulsions in the past 24 hours

D is correct: **two or more convulsions in 24 h** is a defining feature of severe malaria (UCG 2023; WHO). A is wrong: fever alone is uncomplicated (UCG lists hyperpyrexia >39.5 °C only as a supporting feature). B is wrong: severe vomiting means vomiting everything or unable to drink; this child can drink. C is wrong: splenomegaly is common in uncomplicated malaria. E is wrong: severe malarial anaemia is Hb **<5 g/dL**; 7.5 g/dL is moderate anaemia.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Convulsions and coma in a 3-year-old

You are the intern at Mengo Hospital paediatric emergency at 21:00. A 3-year-old boy from Mukono is carried in by his mother. He has had fever for 3 days, was given 'some tablets' from a drug shop yesterday, and had three generalised convulsions today, the last 30 minutes ago. He has not passed urine since morning. Weight 12 kg.

1. What do you do in the first 5 minutes?

> **ETAT**: airway (suction, recovery position), breathing (RR 48, SpO2 93% → oxygen), circulation (HR 158, CRT 2 s, warm hands: not shocked), **coma check**: Blantyre score = eyes 0, motor 1 (withdraws), verbal 1 (moans) = **2**. Not convulsing now. IV access; bedside **glucose, HemoCue Hb, RDT**; cross-match and blood culture. Weigh (12 kg).

2. Glucose is 1.8 mmol/L, Hb 6.4 g/dL, RDT positive. What do you give, with doses?

> **10% dextrose 5 mL/kg = 60 mL IV** over 5–7 min (12 mL of 50% + 48 mL water), recheck glucose at 15–30 min. **IV artesunate 3 mg/kg = 36 mg** (under 20 kg): 60 mg vial + 1 mL bicarbonate + 5 mL saline = 10 mg/mL → **3.6 mL IV** over 5 min; repeat at 12 and 24 h, then daily. Paracetamol 15 mg/kg = 180 mg rectally for fever. No transfusion yet (Hb >6 and no respiratory distress or shock), but keep blood cross-matched.

3. What else must you take from the history now?

> **What were the tablets** (antimalarial? quinine within 12–24 h?); number and type of seizures (focal → meningitis/encephalitis); vomiting, ability to drink; **dark urine**; previous admissions or transfusions (SCD?); **HIV status** of child and mother; **immunisation card** (Hib, PCV for meningitis risk; malaria vaccine); feeding and weight trend; net use; recent travel.

4. One hour later glucose is 5.6 mmol/L, no further seizures, but Blantyre is still 2. Neck is not stiff; pupils equal and reactive; no posturing; BP and pulse stable; platelets 98 x 10^9/L. What is your differential and next step?

> Differential: **cerebral malaria**, **bacterial meningitis** (co-existing), encephalitis, ongoing **subtle seizures**, post-ictal state from an unwitnessed seizure. No signs of raised ICP and platelets adequate → **lumbar puncture** now. Start **ceftriaxone 100 mg/kg = 1.2 g IV** once daily until CSF results are back (also covers bacteraemia). Look for subtle seizures (eye deviation, nystagmus, twitching).

5. CSF is clear: WBC 2/µL, protein 0.3 g/L, glucose 3.2 mmol/L (blood 5.4), Gram stain negative. How do you manage him on the ward?

> **Cerebral malaria** confirmed (normal CSF). Continue artesunate; stop ceftriaxone if blood culture negative and CSF normal (or complete a short course if very sick, per senior). **Nursing**: recovery position, turn 2-hourly, **NGT** (feeds and drugs), eye care, catheter for urine output. **Monitoring**: Blantyre, vitals and **glucose 4-hourly**, urine output, liver size, Hb daily. **Fluids**: maintenance 1100 mL/day (about 46 mL/h) including feeds; no boluses. Treat seizures with diazepam then phenobarbital 15 mg/kg = 180 mg.

6. He wakes on day 2 and takes oral medication. What is your discharge and follow-up plan?

> After at least 3 artesunate doses (24 h) and able to swallow: **DHA–piperaquine** 3 days (UCG), then **monthly DP for 3 months (PDMC)**. **Neurological assessment** before discharge (gait, vision, hearing, speech, behaviour): 10–20% have sequelae; refer for rehabilitation if any. **Hb at about 2–4 weeks** (delayed haemolysis). HIV test if not done; iron and deworming once infection settles; net counselling; danger signs; catch-up malaria vaccine per district policy.

> **Take-away:** In a comatose child with a positive RDT, correct glucose and stop seizures first, give artesunate at the right weight-based dose (3 mg/kg under 20 kg), and do not label it cerebral malaria until meningitis is excluded.

Case 2: Severe malarial anaemia at a HC IV

You are the medical officer at a HC IV in Kasese district. A 14-month-old girl is brought in by her grandmother with 5 days of fever and 'fast breathing' since this morning. She is still breastfeeding. Weight 8 kg, length 74 cm, MUAC 12.9 cm. The HC IV has a small blood fridge but only one unit of group O whole blood today.

1. What examination findings will you look for and how do you triage her?

> ETAT: RR 66, **deep breathing** without chest indrawing, SpO2 94%, HR 172, CRT 2 s, warm hands. **Severe palmar pallor**, no jaundice. Blantyre 5 but irritable; spleen 4 cm; liver 2 cm (baseline). Chest: no crackles. No oedema of the feet. **Emergency/priority**: severe respiratory distress + severe pallor → resuscitation room. Weight-for-length Z about −1 (no SAM; MUAC ≥12.5 cm).

2. Bedside tests: RDT positive, HemoCue Hb 3.2 g/dL, glucose 4.9 mmol/L. What is your diagnosis and immediate management?

> **Severe malaria with severe malarial anaemia (Hb <5) and respiratory distress from metabolic acidosis.** Oxygen if SpO2 <90% or severe distress. **IV artesunate 3 mg/kg = 24 mg**. **Transfuse whole blood 20 mL/kg = 160 mL over 3–4 h** (Hb ≤4 = absolute indication). No saline bolus (not shocked; FEAST). Group, cross-match and check blood safety labelling; take blood for blood culture/FBC before transfusion if possible.

3. What will you monitor during the transfusion?

> Temperature, pulse, RR and BP **at baseline, 15 minutes, then every 30 min**; watch for **reactions** (fever, rash, rigors, wheeze, hypotension → stop) and **overload** (rising RR, crackles, **liver enlarging**, gallop) → slow/stop and give **furosemide 1 mg/kg = 8 mg IV**. Glucose before and after. Count the transfused volume in the daily fluid total.

4. After the transfusion her breathing is 38/min and quiet, Hb 7.0 g/dL. The next morning she is feeding. Which other problems will you look for?

> **Bacteraemia** (NTS is common with malarial anaemia: give ceftriaxone if febrile despite artesunate, high WBC or very ill), **sickle cell disease** (sickling test: splenomegaly at this age is also seen with malaria, but recurrent severe anaemia or a positive family history suggests SCD), **HIV** (test child; mother's status), **iron deficiency and hookworm** (too young for much hookworm, but check diet), and **growth/feeding** (complementary feeding counselling).

5. Plan her treatment completion, discharge and prevention.

> Artesunate at 0, 12, 24 h then switch to **DP for 3 days**; **PDMC**: monthly DP for 3 months (both UCG 2023 and WHO, since she had severe anaemia). **Hb recheck at 2 and 4 weeks**; start **iron** (elemental iron about 3 mg/kg/day) once acute illness settles, with folic acid; deworm from 1 year (albendazole 200 mg at 12–23 months). LLIN for the household; malaria vaccine doses per UNEPI (she is past the 6–8 month window, so follow district catch-up policy). Teach the grandmother danger signs.

6. If there had been no blood at the HC IV, what would you have done?

> Give the first dose of **IV/IM artesunate**, oxygen, keep warm, correct glucose, then **refer urgently** to the RRH (Bwera/Fort Portal) with a referral note, a health worker escort if possible, the mother/grandmother to breastfeed, and call ahead. Do **not** give fluid boluses to 'buy time'.

Take-away: Deep breathing plus profound pallor in a child with malaria is acidosis from anaemia: the treatment is artesunate plus 20 mL/kg whole blood, not a fluid bolus, and severe anaemia earns post-discharge chemoprevention and Hb follow-up.

Case 3: From VHT to HC III: getting the referral right

A VHT in Kayunga district sees a 2-year-old girl (weight 11 kg) at 07:00. She has had fever for 2 days, and since last night she vomits everything and will not sit up. The VHT's RDT is positive. The family must travel 15 km to the HC III.

1. What should the VHT do before the child leaves?

> Recognise **danger signs** (vomiting everything, unable to sit = prostration). Give **rectal artesunate 100 mg** (one suppository; 4 months–<3 years, 5–<14 kg), hold the buttocks together for 10 min, repeat if expelled within 30 min. Keep breastfeeding or give sips of sugar water to prevent hypoglycaemia, keep warm, write a referral note with the time of the dose, and **arrange immediate transport**.

2. She arrives at the HC III 3 hours later, drowsy (Blantyre 4), temperature 39.8 °C, RR 40, HR 150, CRT 2 s, glucose 2.3 mmol/L. What do you do?

> Glucose 2.3 is below 2.5 → **10% dextrose 5 mL/kg = 55 mL IV**, recheck. **IV (or IM) artesunate 3 mg/kg = 33 mg** now (rectal dose does not replace it). **Paracetamol 15 mg/kg = 165 mg**. Look for neck stiffness, pallor (HemoCue), dark urine. If the HC III cannot monitor overnight and give the 12-hour dose, refer to the HC IV with the drug chart.

3. At the HC IV next morning she is alert, Hb 8.9 g/dL, tolerating feeds. When can you switch to oral treatment and what will you give?

> After **at least 3 doses (0, 12, 24 h)** of parenteral artesunate and once she tolerates oral intake: a **full 3-day ACT** (UCG: discharge severe malaria on **DHA–piperaquine**: 10–20 kg band → 1 tablet daily × 3 days), then PDMC monthly × 3 months.

4. Her mother asks why the child got sick despite having a net. What will you assess and advise?

> Check **net use** (every night? torn? enough nets for all children?), the **immunisation card** (malaria vaccine at 6, 7, 8 and 18 months: she is 2 years; check whether she received any doses, as the rollout began in April 2025), early care-seeking (go to the VHT/HC the same day as fever), IRS in the area, and drain stagnant water. Nets reduce but do not abolish risk.

5. What would make this case a 'treatment failure' and how would you confirm it?

> Parasites **persisting or returning within 28 days** of a full, correctly taken course, confirmed by **microscopy** (not RDT, which stays positive). A child who is still febrile with a rising or unchanged parasite count on day 3 suggests slow clearance (relevant given **kelch13 partial resistance in northern Uganda**); report to the senior and use an alternative ACT.

Take-away: Rectal artesunate at the VHT or HC II is only the start: the child must reach a facility for injectable artesunate, glucose and fluids, followed by a full ACT and chemoprevention.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.